

 UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF AS	MIDTERM EXAM		Semester/ Academic year		213	2021 - 2022
			Date	10 July 2022		
	Course title		Calculus 2			
	Course ID		MT1005			
Duration		50 mins	Question sheet code: 1764			
-This is a closed book exam. Total available score: 10 points. -At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet.						
Student's full name: Student ID: Invigilator 1: Invigilator 2:						

There are 20 questions. Each question is assigned 0.5 point. (L.O.1)

Q. 1. Let g be the function defined by

$$g(x, y, z) = \left(\frac{3x^3 - xyz^4 - 2xyz}{(x^2 + y^4 + z^6)^8} \right)^{10} \ln \left(\frac{1}{e^{x-(y-2)^2-z^3}} \right).$$

Evaluate $g_y(-1, 0, 0)$.

- ☐ (A) -4245690 ☐ (B) -3753892 ☐ (C) 1845654 ☐ (D) -4194304
☐ (E) -2344900

Q. 2. Choose a point in the domain of the function $f(x, y) = \ln \left(\sin \left(\frac{1}{x^2 + y^2} \right) \right)$ from the following.

- ☐ (A) $M(-0.3, -0.3)$ ☐ (B) $M(0.1, -0.3)$ ☐ (C) $M(0.2, -0.2)$
☐ (D) $M(0.2, -0.3)$ ☐ (E) $M(0.5, -0.2)$

Q. 3. Let M be a point on the circle $(x+2)^2 + y^2 = 4$. Given that M has polar coordinates $(r_0, -\frac{3\pi}{4})$ (in the standard polar coordinate system $x = r \cos \phi, y = r \sin \phi$). What is the value of r_0 ?

- ☐ (A) $3\sqrt{2}$ ☐ (B) $2\sqrt{2}$ ☐ (C) 4 ☐ (D) 2
☐ (E) $\sqrt{2}$

Q. 4. Let (S) be the paraboloid $z = 10 - (x-1)^2 - (y-2)^2$, and let M be the highest point (relative to the plane Oxy) on (S) . Let (P) be the tangent plane to (S) at M . Which point from the following is on the plane (P) ?

- ☐ (A) $N(1, 2, 9)$ ☐ (B) $N(-1, 2, 9)$ ☐ (C) $N(-1, -2, 9)$
☐ (D) $N(1, -2, -9)$ ☐ (E) $N(3, 4, 10)$

Q. 5. Let $u(x, y)$ be a function such that

$$u_x(-1, 0) = -4, u_y(-1, 0) = -2, u(-1, 0) = 3.$$

Consider the function $F(x, y) := \cos(u(x, y))$. Evaluate $F_x(-1, 0)$.

- (A) $\cos(4)$ (B) $4 \sin(3)$ (C) $6 \sin(3)$
(D) $4 \sin(3) + \cos(4)$ (E) $6 \sin(3) + \cos(4)$

Q. 6. Let M be a point different from the origin and on the intersection between the line $y - x = 0$ and the parabola $x - y^2 = 0$. Which one from the following is the polar coordinates of M (in the standard polar coordinate system $x = r \cos \phi, y = r \sin \phi$)?

- (A) $M(1, \frac{\pi}{3})$ (B) $M(\sqrt{2}, \frac{\pi}{4})$ (C) $M(1, -\frac{\pi}{4})$ (D) $M(\sqrt{2}, -\frac{\pi}{4})$
(E) $M(1, \frac{\pi}{4})$

Q. 7. Which number from the following is in the range of the function $f(x, y) = e^{x^2 - y^2}$?

- (A) -2.71 (B) 0 (C) -0.3 (D) 0.2
(E) -0.4

Q. 8. Let $D = \{(x, y) : x^2 + y^2 \leq 1, y \leq -x\}$. In the polar coordinate system $x = r \cos(\phi), y = r \sin(\phi)$, what is the range for (r, ϕ) corresponding to the domain D ?

Choose the correct statement from the following.

- (A) $0 \leq r \leq 1, -\pi \leq \phi \leq \pi$
(B) $0 \leq r \leq 1, -\frac{\pi}{4} \leq \phi \leq \frac{3\pi}{4}$
(C) $0 \leq r \leq 1, -\frac{5\pi}{4} \leq \phi \leq -\frac{\pi}{4}$
(D) $0 \leq r \leq 1, -\frac{\pi}{4} \leq \phi \leq \frac{\pi}{4}$
(E) $0 \leq r \leq 1, \frac{3\pi}{4} \leq \phi \leq \frac{3\pi}{2}$

Q. 9. In $\mathbb{R}^3$, let (P) be the plane of equation $4x - 2y - 2z + 8 = 0$. Which function in the following whose graph is the plane (P) ?

- (A) $f(x, y) = 8x - 4y + 4$
(B) $f(x, y) = 2x - 3y + 8$
(C) $f(x, y) = x - y + 2$
(D) $f(x, y) = 2x - 2y + 1$
(E) $f(x, y) = 2x - y + 4$

Q. 10. In $\mathbb{R}^2$, what is the greatest distance from a point on the circle $x^2 + y^2 = 1$ to the point $M(-1, 7)$?

- (A) $\sqrt{51 + 10\sqrt{2}}$ (B) $\sqrt{50 + \sqrt{53}}$ (C) $\sqrt{50 + 18\sqrt{2}}$
(D) $\sqrt{51 + \sqrt{53}}$ (E) $\sqrt{51}$

Q. 11. Find the maximum value of the function $f(x, y) = 13 - x^2 - y^2$ for (x, y) in the domain

$$\{(x, y) : x^4 + y^6 \leq 1\}.$$

- (A) $14\sqrt{2}$ (B) 12 (C) 11 (D) $12\sqrt{2}$
(E) 13

Q. 12. The points given in the following are represented in polar coordinates. Which point is inside the circle $x^2 + y^2 = 4$?

- ☐ (A) $(5, -\frac{\pi}{6})$
☐ (B) $(\frac{3}{2}, \frac{\pi}{6})$
☐ (C) $(5, \frac{\pi}{3})$
☐ (D) $(3, \frac{\pi}{2})$
☐ (E) $(6, \frac{\pi}{4})$

Q. 13. How many saddle points does the function $f(x, y) = 2022 + 2021ye^{-x}$ have?

- ☐ (A) 0
 ☐ (B) 1
 ☐ (C) 2
 ☐ (D) 3
☐ (E) Infinitely many

Q. 14. Among the following directions, which one that the function $f(x, y) = \frac{1}{x^2+y^4} \sin(x^2 + y^4)$ goes uphill most rapidly at $M(-1, 1)$?

- ☐ (A) $\vec{u}(1, 0)$
☐ (B) $\vec{u}(-2, 5)$
☐ (C) $\vec{u}(1, -2)$
☐ (D) $\vec{u}(3, -1)$
☐ (E) $\vec{u}(4, -6)$

Q. 15. Evaluate the double integral

$$\iint_D x^2 dA$$

where D is the disk $\{(x, y) : x^2 + y^2 \leq 9\}$.

- ☐ (A) 9π
☐ (B) $\frac{9\pi}{2}$
☐ (C) $\frac{81\pi}{4}$
☐ (D) $\frac{9\pi}{4}$
☐ (E) $\frac{27\pi}{4}$

Q. 16. Let (C) be the intersection between the surface $z = \sqrt{8 - x^2 - y^2}$ and the cone $z = \sqrt{x^2 + y^2}$. The projection of (C) onto the plane Oxy is a circle. Evaluate its radius.

- ☐ (A) 1
 ☐ (B) 2
 ☐ (C) 4
 ☐ (D) $2\sqrt{2}$
☐ (E) $2\sqrt{3}$

Q. 17. Evaluate the value of the function

$$g(x, y, z) = \left(\frac{3x^3 - xyz^4 - 2^{xyz}}{(x^2 + y^4 + z^6)^8 + 1} \right)^{10} \ln \left(\frac{1}{e^{x-(y-2)^2-z^3}} \right)$$

at $M(x = 0, y = 2, z = 0)$.

- ☐ (A) $-2 \ln(2)$
☐ (B) 0
 ☐ (C) $3 \ln(2)$
☐ (D) $2 \ln(2)$
☐ (E) $-3 \ln(2)$

Q. 18. In $\mathbb{R}^3$, let (C) be the intersection between the cylinder $x^2 + y^2 = 5$ and the plane $2x + 3y - 4z + 8 = 0$. Which point from the following is on the curve (C) ?

- ☐ (A) $M(1, 2, -1)$
☐ (B) $M(1, 2, 1)$
☐ (C) $M(-1, 2, 1)$
☐ (D) $M(1, -2, 1)$
☐ (E) $M(-1, -2, 1)$

Q. 19. Which function from the following has a local minimum at the point $M(1, 0)$?

- ☐ (A) $f(x, y) = (x - 1)^2 + y^2 + 3$
☐ (B) $f(x, y) = -(x - 1)^2 - y^2 + 3$
☐ (C) $f(x, y) = (x - 1)y + 3$
☐ (D) $f(x, y) = (x - 1)y^2 + 3$
☐ (E) $f(x, y) = (x - 1)^2 y + 3$

Q. 20. Evaluate the double integral

$$\iint_D 3dA$$

where D is the triangular region with vertices: $A(-3, 0)$, $B(0, 5)$ and $C(7, 0)$.

(A) 150

(B) 25

(C) 50

(D) 30

(E) 75